

The diagrammatic equation is as follows:

$$\begin{aligned}
 & \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} = \text{disc}_{p_1^2} \text{disc}_{p_2^2}
 \end{aligned}$$

Detailed description of the diagrams:

- Diagram 1:** Two vertices on the left and right, each with two external lines labeled p_1 and p_2 . A wavy line connects the two vertices, passing through a red vertical line. Blue and orange shaded regions are present on the external lines.
- Diagram 2:** Similar to Diagram 1, but with different internal line connections and momentum labels k and $k - p_1$.
- Diagram 3:** Similar to Diagram 1, but with different internal line connections and momentum labels k and $k - p_2$.
- Diagram 4:** Similar to Diagram 1, but with different internal line connections and momentum labels k and $k - p_1$.
- Right-hand side:** The product of two disc operators, $\text{disc}_{p_1^2}$ and $\text{disc}_{p_2^2}$, represented by two separate diagrams with wavy lines and external lines.